1. X = No. of defective bulbs in box

$X \sim$ binomial $(n = 5,\ p = 0.10)$

$P(X \geq 1)$

$= 1 - P(X = 0) = \cdots = 0.4095$

2. $np = 4 \quad \cdots\cdots (A)$

$np(1-p) = 2 \quad \cdots (B)$

$(B) \div (A), \quad 1 - p = 0.5$

$\therefore \quad p = 0.5$

$\therefore \quad n = \dfrac{4}{0.5} = 8$

$P(X = 0) = \cdots = 0.0039$

3. X = No. of misprints in a page

$X \sim$ Poisson $(\lambda = 3)$

$$P(X = 2) = \frac{e^{-\lambda} \lambda^{x}}{x!}$$

$$= \frac{e^{-3}\, 3^{2}}{2!}$$

$$= 0.2240$$

2

4. $\lambda_1 = 3$

$\lambda_2 = 2$

$\lambda = \lambda_1 + \lambda_2 = 3 + 2 = 5$

X = No. of people between 9:00 - 11:00.

$X \sim \text{Poisson}(\lambda = 5)$

$$P(X \leq 2) = P(0) + P(1) + P(2)$$
$$= \ldots$$
$$= 0.125$$

5. Part 4 (solution done there)

6. X = No. of good phones out of 5

$(X = 0, 1, 2, 3, 4, 5)$

$P(X \geq 2)$ Hypergeometric

$$= 1 - P(0) - P(1)$$

$$= 1 - \frac{{}^{20}C_0 \times {}^{10}C_5}{{}^{30}C_5} - \frac{{}^{20}C_1 \times {}^{10}C_4}{{}^{30}C_5}$$

$$= 0.969$$

7. Large population, $X =$ No. of good

$X \sim$ binomial $\left(n=5,\ p=\frac{2000}{3000}=\frac{2}{3}\right)$

$P(X \geq 2)$

$= 1 - P(0) - P(1)$

$= 0.955$

8. $X =$ No. of failures before 3 successes

$X \sim$ negative binomial $\left(r=3,\ p=\frac{1}{6}\right)$

$P(X=2)$

$$= \binom{r+x-1}{r-1} p^r (1-p)^x$$

$$= \binom{3+2-1}{3-1} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^2$$

$= 0.019$